

Smart Room Occupancy Counter

Problem Statement

In smart buildings, classrooms, offices and large public events such as cultural fests and exhibitions, monitoring the number of people inside a room or enclosed area is essential for energy management, safety and capacity control. Overcrowding in confined spaces can lead to safety risks, discomfort, and inefficient use of resources such as lighting, ventilation and air-conditioning. Most existing visitor counting systems are implemented using microcontrollers and software-based processing. These solutions suffer from sequential processing delays, timing uncertainty, higher power consumption and unreliable detection during fast or continuous movement, making them unsuitable for real-time safety-critical applications. To address these limitations, there is a need for a low-power, real-time, hardware-based occupancy monitoring system capable of accurately counting people entering and exiting through a doorway and ensuring that the number of occupants does not exceed a predefined capacity.

Solution

The proposed solution is to design a bidirectional room occupancy monitoring system using hardware logic written in Verilog HDL and implement it on the PYNQ-Z2 FPGA as a reconfigurable prototyping platform. Two IR break-beam sensors are placed at the doorway to detect human movement during entry and exit, and a sensor interface module inside the FPGA performs debouncing, synchronization and edge detection to ensure reliable signal processing. A Finite State Machine (FSM) analyses the sequence of sensor activation to determine the direction of movement, after which a real-time hardware counter updates the room occupancy by incrementing for entry and decrementing for exit events. The system also includes capacity monitoring to generate alerts when occupancy reaches a predefined limit and supports energy management by enabling automatic control of lighting and ventilation based on room usage. The complete design is validated on FPGA hardware to ensure correct real-time operation, making the architecture ready for future migration into a custom chip for large-scale deployment.

Literature Survey

Existing research on room occupancy and visitor counting systems mainly focuses on low-cost sensing and real-time monitoring for smart buildings and public safety. Early systems used simple infrared (IR) beam sensors with microcontrollers to detect entry and exit by analysing the sequence of sensor activation. Several studies proposed bidirectional visitor counters using Arduino, 8051 and ESP32 platforms for classrooms and offices, highlighting ease of implementation and IoT connectivity. Later works extended these systems to energy management by automatically controlling lighting and ventilation based on occupancy levels. Camera-based approaches using computer vision and deep learning were also explored to improve accuracy in crowded environments, but they introduce higher complexity and power

consumption. Researchers have emphasized the need for reliable real-time systems for applications such as cultural events, exam halls and smart campuses where capacity control is critical. However, most existing solution.

SDG Goals Aligned:

SDG 11 — Sustainable Cities & Communities

- 1.Prevents overcrowding
- 2.Improves safety in public events

SDG 9 — Industry, Innovation & Infrastructure

1. Can be converted to ASIC for deployment

TRL 5 - Prototype validation (FPGA prototype, Real sensors)



Market Analysis:

The demand for people counting and occupancy monitoring systems is rapidly growing due to the expansion of smart buildings, shopping malls, transport hubs, offices and large public events where real-time entry–exit monitoring is essential for **safety and energy management**. According to market research, the global people counting system market was valued at about **1.26 billion in 2024** and is expected to reach around **2.65 billion by 2030**, showing strong global adoption of such systems. The growth is driven by increasing safety requirements in airports, train stations, retail stores and event venues, where real-time occupancy monitoring is required for crowd control and building management. Studies also highlight that these systems are widely used in hotels, shopping malls and hospitality sectors to monitor visitor flow, ensure safety regulations and optimize energy usage. Additionally, smart-city initiatives and post-pandemic occupancy regulations have further increased adoption, with the market projected to reach nearly USD 2.99 billion by 2031. Many sectors such as transportation hubs, +retail chains and large event venues require large-scale deployment of entry–exit monitoring systems across thousands of locations, demonstrating the need for scalable hardware solutions like the proposed FPGA-based system. Hence, the growing global market and wide industrial adoption clearly justify the relevance and demand for a real-time occupancy monitoring system.

[People Counting System Market – Grand View Research](#)

[People Counting System Market – Markets and Markets](#)

[People Counting System Market – Consegic Business Intelligence](#)

[People Counting System Market – Mordor Intelligence](#)

[Research and Markets Report](#)

Empathy Mapping

Purpose

Empathy mapping is used to understand the needs, expectations, and challenges of stakeholders who will interact with or benefit from the occupancy monitoring system. This analysis ensures that the engineering solution aligns with real-world requirements rather than only technical assumptions.

Target Stakeholders

- Facility Managers
- Institutional Administrators
- Event Coordinators
- Safety Compliance Officers

Empathy Analysis

- Accurate real-time occupancy data is required.
 - Overcrowding must be prevented.
 - Manual counting is unreliable.
 - The system must be reliable during continuous or fast movement.
 - The solution should operate without delay.
 - It must integrate with existing infrastructure.
 - It should be scalable to multiple rooms.
 - Sequential processing delays in software systems.
 - High power consumption in camera-based systems.
 - Privacy risks due to image capture.
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Modern Engineering Perspective

Hardware Acceleration and Determinism - Unlike microcontroller-based systems that rely on sequential instruction execution, the FPGA implementation provides: Parallel processing, Hardware-level state machine control

Power-Efficient Design - By leveraging digital hardware logic, the system consumes significantly less power.

Privacy and Ethical Engineering - The system avoids image acquisition or personal data storage. IR break-beam sensing ensures. No biometric data collection, No video recording

FPGA-to-ASIC Scalability

The system follows a professional hardware development workflow:

1. RTL design in Verilog HDL
2. Functional simulation and verification
3. FPGA prototyping on PYNQ-Z2
4. Hardware validation
5. ASIC migration

Minimum Viable Product (MVP)

A Minimum Viable Product (MVP) is the simplest functional version of the system that demonstrates the core value proposition while minimizing development complexity.

For this project, the MVP validates accurate bidirectional occupancy counting using hardware logic.

Core Functional Scope

The MVP includes:

- Two IR break-beam sensors for entry/exit detection
- Sensor synchronization and signal conditioning module
- Finite State Machine (FSM) for direction analysis
- Real-time occupancy counter
- Reset functionality

Core Value Demonstration

The MVP successfully proves:

- Correct direction detection (A→B for entry, B→A for exit)
- Instantaneous counter updates
- Immunity to noise through debouncing and synchronization
- Stable operation under continuous movement

Component	Estimated Price (₹)	Product / Link
FPGA Board — PYNQ-Z2	~₹20,000 – ₹26,000	(Academic board — supplier specific, Amazon may not list it) – example: search “PYNQ-Z2 board India”
IR Break-Beam Sensors (2 pcs)	~₹50	https://ainow.in/product/ir-sensor-for-smart-car/

Reference Links

[Vision-based occupancy detection in indoor environments: A comparison of standard RGB and thermal cameras - ScienceDirect](#)

[A Computer vision based system for human detection and automatic people counting](#)

[Anyone here? Smart Embedded Low-Resolution Omnidirectional Video Sensor to Measure Room Occupancy | IEEE Conference Publication | IEEE Xplore](#)